SOLUTION
SECTION-I

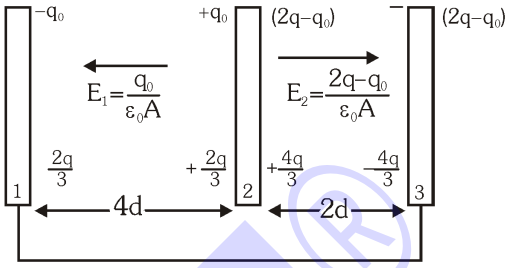
Single Correct Answer Type **8 Q. [3 M (–1)]**

1. Ans. (B)

Sol. $E_1 4d = E_2\, 2d$

$$\frac{q_0}{\epsilon_0 A}4d = \frac{(2q-q_0)}{\epsilon_0 A}2d \Rightarrow q_0 = \frac{2q}{3}$$

$$V_{23} = \frac{4q}{3\epsilon_0 A}\times 2d = \frac{8qd}{3\epsilon_0 A} \ldots\ldots (i)$$



2. Ans. (C)

Sol. Apply KVL

$$\varepsilon = \frac{q}{C} - iR + \frac{RC\varepsilon - q}{C} = 0$$

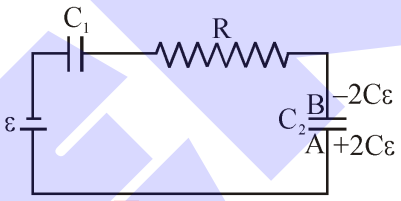
$$\frac{3C\varepsilon - 2q}{C} = iR$$

$$i = \frac{dq}{dt}$$

$$\frac{dq}{dt} = \frac{3C\varepsilon - 2q}{C}$$

$$\int_0^q \frac{dq}{3C\varepsilon - 2q} = \frac{1}{RC}$$

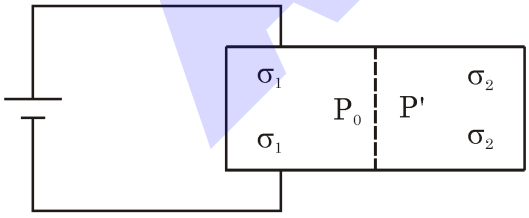
$$q = \frac{3C\varepsilon}{2}(1 - e^{-1/RC})$$



On plate B, Charge is $-2C\varepsilon + q = \frac{-C\varepsilon}{2}(3 + 3e^{-1/RC})$

3. Ans. (C)

Sol.



$$E = \frac{V}{d}$$

$\sigma_1 \neq \sigma_2$

But

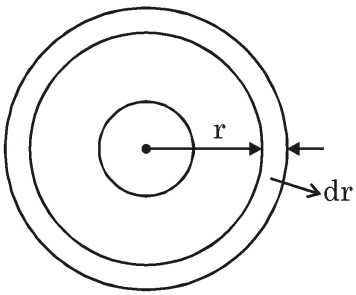
$$E = \frac{V}{d}$$

So electric field at P and P' is equal.

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4. Ans. (B)

Sol.



$$\text{Resistance} = R = \int_R^\infty \frac{\rho dr}{4\pi r^2}$$

$$R_{es} = \frac{\rho}{4\pi}\int_R^\infty \frac{dr}{r^2}$$

$$R_{es} = \frac{\rho}{4\pi}\int_R^\infty r^{-2}dr$$

$$R_{es} = \frac{\rho}{4\pi} - \left[\left(\frac{1}{r}\right)\right]_R^\infty = \frac{-\rho}{4\pi}\left[\frac{1}{\infty} - \frac{1}{R}\right]$$

$$R_{es} = \frac{\rho}{4\pi R}$$

For discharging circuit

$$Q = Q_0 e^{-\frac{t}{R_{es}C}}$$

$$\frac{Q_0}{2} = Q_0 e^{-\frac{t}{R_{es}C}}$$

$$\frac{1}{2} = e^{-\frac{t}{R_{es}C}}$$

$$-\ell n2 = -\frac{t}{R_{es}C}$$

$$t = R_{es}C\ell n2$$

$$t = \frac{\rho}{4\pi R} \times k4\pi\varepsilon_0 R\ell n2$$

$$t = \rho k\varepsilon_0 \ell n2$$

5. Ans. (A)

Sol. $E \times \frac{1}{k} \Rightarrow \Delta V \times \frac{1}{k}$

$\Rightarrow E_3 < E_2 < E_1$ and $\Delta V_3 < \Delta V_2 < \Delta V_1$

6. Ans. (A)

Sol. we know $q' = -q\left(1 - \frac{1}{k}\right)$,

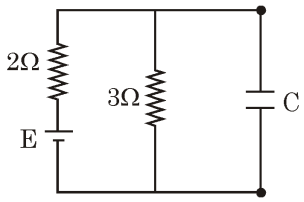
Therefore, $|q'| < |q|$

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7. Ans. (A)

Sol.



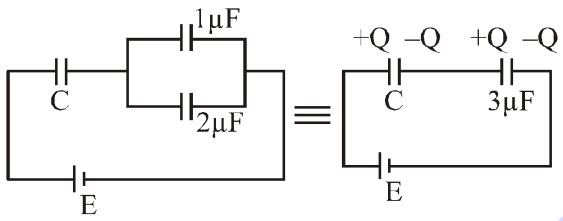
$q_1 = CE$, when switch is closed.

$$E_{eq} = \frac{3E}{5}$$

$$q_2 = CE \times \frac{3}{5}$$

8. Ans. (D)

Sol.



$$C_{eq} = \frac{3C}{3+C} \qquad Q = Ceq \cdot E$$

$$Q_2 = \frac{2CE}{(C+3)} = \frac{3CE}{(C+3)}$$

$$= 2E\left[1-\frac{3}{C+3}\right]$$

$$Q_{2/\text{when } C = 1\mu F} = 2E\left[\frac{1}{4}\right] = \frac{E}{2}$$

$$Q_{2/\text{when } C = 3\mu F} = 2E\left[\frac{1}{2}\right] = E$$

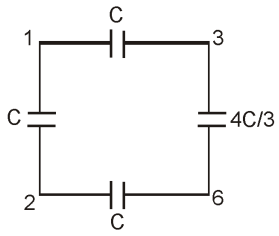
$$\frac{dQ}{dC} = -\frac{2E \cdot 3}{(C+3)}(-1) = \frac{6E}{(C+3)^2} > 0$$

$$\frac{d^2Q}{dC^2} = \frac{6E(-2)}{(C+3)^3} = -\frac{12E}{(C+3)^2} < 0$$

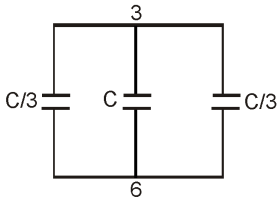
Multiple Correct Answer Type **7 Q. [4 M (–1)]**

9. Ans. (A,B,C)

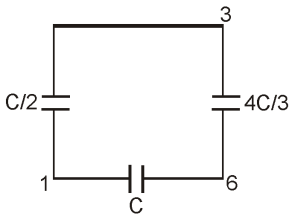
Sol. (A)



(B)



(C)

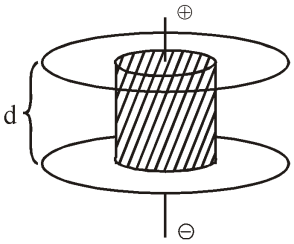


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10. Ans. (B,D)

11. Ans. (A,C)

Sol.



$C = C_{di} + C_{empty}$

$= \frac{k \in_0 (\pi R^2)}{d} + \frac{\in_0 \pi (3R^2)}{d}$

$C = \frac{(k+3) \in_0 \pi R^2}{d}$

12. Ans. (B,D)

13. Ans. (A, B, C, D)

Sol. $V_0 = I_0 R = 10 \times 10 = 100$ volts (since, $I_0 = 10$ amp from figure) Hence (A) is correct

Also: $I = I_0 e^{-t/RC}$

Taking log; $\ln\left(\frac{I_0}{I}\right) = \frac{t}{RC} \Rightarrow C = \frac{t}{R/n\,(I_0/I)}$

At; $t = 2$ sec, $I = 2.5$ A

$C = \frac{2}{10 \ln\left(\frac{10}{25}\right)}$

$C = \frac{2}{10 \ln 4} = \frac{2}{10 \times 2 \ln 2} = \frac{1}{10 \ln 2}$ F Hence (B) is correct.

$C = \frac{1}{10 \ln 2}$.

Heat produced $= \frac{1}{2} CV^2 = \frac{1}{2}\left(\frac{1}{10 \ln 2}\right)(100^2) = \frac{500}{\ln 2}$ joules.

Hence, (C) is correct

Thermal power in the resistor will decrease with a time constant $\frac{1}{2 \ln 2}$ second.

Hence, (D) is correct.

14. Ans. (A, B, C)

15. Ans. (A, C)

Linked Comprehension Type (2 Para × 2Q.) [3 M (-1)]
(Single Correct Answer Type)

16. Ans. (B)

Sol. $\left(i - \frac{dq}{dt}\right) R = \frac{q}{C}$

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$$\Rightarrow \frac{dq}{i-\frac{q}{RC}} = dt$$

$$\& \; q = i\,RC\left(1-e^{-\frac{t}{RC}}\right)$$

17. Ans. (C)

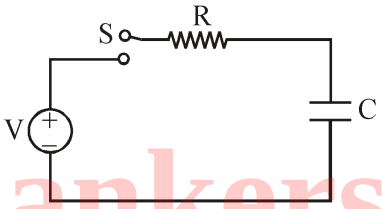
Sol. $$\frac{d}{dt}\left(\frac{q^2}{2C}\right)=\left(i-\frac{dq}{dt}\right)^2 R$$

$$\Rightarrow \frac{q}{C}\frac{dq}{dt}=\left(i-\frac{dq}{dt}\right)\times R\left(i-\frac{dq}{dt}\right)$$

$$\Rightarrow \frac{2dq}{dt} = i \Rightarrow t = RC\,\ell n2$$

18. Ans. (A)

Sol.



When switch is closed for a very long time capacitor will get fully charged & charge on capacitor will be $q = CV$

Energy stored in capacitor $\in_C = \frac{1}{2}CV^2$ ...(i)

Work done by battery $(\omega) = Vq = VCV = CV^2$

dissipated across resistance $\in_D$ = (work done by battery) – (energy store)

$$\in_D = CV^2 - \frac{1}{2}CV^2 = \frac{1}{2}CV^2 \qquad \text{...(ii)}$$

from (i) & (ii)

$\in_D = \in_C$

19. Ans. (A)

Sol. For process (1)

Charge on capacitor = $\frac{CV_0}{3}$

energy stored in capacitor = $\frac{1}{2}C\frac{V_0^2}{9} = \frac{CV_0^2}{18}$

work done by battery = $\frac{CV_0}{3}\times\frac{V}{3} = \frac{CV_0^2}{9}$

Heat loss = $\frac{CV_0^2}{9} - \frac{CV_0^2}{18} = \frac{CV_0^2}{18}$

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For process (2)

Charge on capacitor = $\frac{2CV_0}{3}$

Extra charge flow through battery = $\frac{CV_0}{3}$

Work done by battery : $\frac{CV_0}{3}\cdot\frac{2V_0}{3} = \frac{2CV_0^2}{9}$

Final energy store in capacitor : $\frac{1}{2}C\left(\frac{2V_0}{3}\right)^2 = \frac{4CV_0^2}{18}$

energy store in process 2 : $\frac{4CV_0^2}{18} - \frac{CV_0^2}{18} = \frac{3CV_0^2}{18}$

Heat loss in process (2) = work done by battery in process (2) – energy store in capacitor process (2)

$= \frac{2CV_0^2}{9} - \frac{3CV_0^2}{18} = \frac{CV_0^2}{18}$

For process (3)

Charge on capacitor = CV_0

extra charge flow through battery : $CV_0 - \frac{2CV_0}{3} = \frac{CV_0}{3}$

work done by battery in this process : $\left(\frac{CV_0}{3}\right)(V_0) = \frac{CV_0^2}{3}$

find energy store in capacitor : $\frac{1}{2}CV_0^2$

energy stored in this process : $\frac{1}{2}CV_0^2 - \frac{4CV_0^2}{18} = \frac{5CV_0^2}{18}$

heat loss in process (3) : $\frac{CV_0^2}{3} - \frac{5CV_0^2}{18} = \frac{CV_0^2}{18}$

Now total heat loss (E_D) : $\frac{CV_0^2}{18} + \frac{CV_0^2}{18} + \frac{CV_0^2}{18} = \frac{CV_0^2}{6}$

final energy store in capacitor : $\frac{1}{2}CV_0^2$

so we can say that $E_D = \frac{1}{3}\left(\frac{1}{2}CV_0^2\right)$

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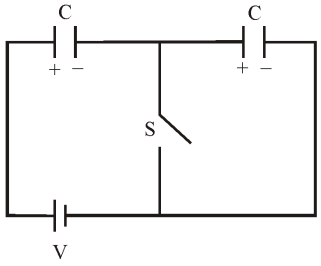
SECTION-II

Numerical Answer Type Question **1Q.[3M(0)]**

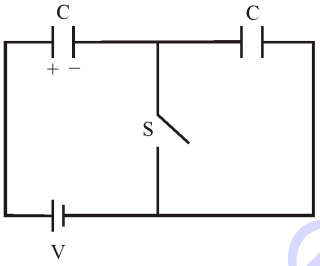
(upto second decimal place)

20. Ans. 4.50

Sol.



initial



final

$$U_i = \frac{1}{2}C.\left(\frac{V}{2}\right)^2.2$$

$$= \frac{CV^2}{4}$$

$$U_f = \frac{1}{2}CV^2$$

$$W_{Cell} = V.\Delta q = V.\left(CV - \frac{CV}{2}\right) = \frac{CV^2}{2}$$

Now $U_i + W = U_f + H$

$$\therefore \quad H = \frac{CV^2}{4} = \frac{2\times 9}{4} = 4.50$$

SECTION-III

Numerical Grid Type (Ranging from 0 to 9) **2 Q. [4 M (0)]**

21. Ans. 4

Sol. $\left(\frac{Q^2}{2\epsilon_0 A}\right) \times x = \frac{1}{2}Kx^2$

$$\Rightarrow x = \frac{Q^2}{\epsilon_0 AK}$$

22. Ans. 7

Sol. $F = -\frac{dU}{dx}$ & $U = \frac{Q^2}{2C}$

Where, $C = \frac{\epsilon_0\, \ell(\ell - x)}{d} + \frac{\epsilon_r \epsilon_0\, \ell(x)}{d}$

SECTION-IV

Matrix Match Type (4 × 5) **3 Q. [8 M (for each entry +2(0)]**

23. Ans. (A) → **(Q, R, S, T)** ; **(B) → (P, S)** ; **(C) → (R, T)** ; **(D) → (P, R, T)**

Sol. (A) Motion is along force then work done is positive

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(B) P–iniitial $\frac{E_0}{K}$, final E_0

S–initial $\frac{v_0}{d}$ decreases so E increase

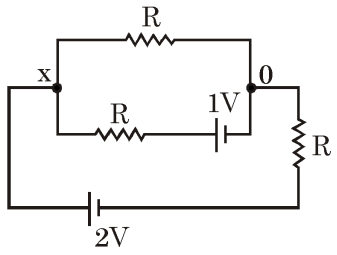
(C) Charge remain same capacitance increases $\frac{Q^2}{2C}$

(D) $F = \frac{Q}{2A\epsilon_0}Q$

where battery is not connected

24. Ans. (A) Q,R,T; (B) P,S; (C) R,S; (D) P,S

Sol. For A

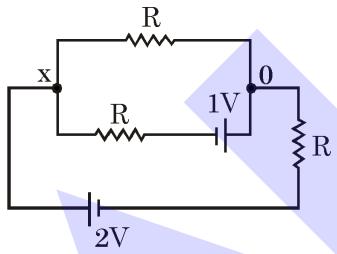


By nodal analysis

$x + x - 1 + x - 2 = 0$

$x = 1\ V,\ I_2 = 0, I_1 = 1A$

For B



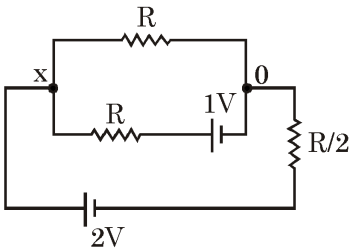
By nodal analysis

$x + x + 1 + x - 2 = 0$

$x = \frac{1}{3}\ V,$

$I_2 = \frac{4}{3}A, I_1 = \frac{5}{3}A$

For C



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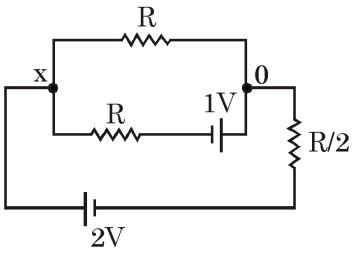
By nodal analysis

$x + x - 1 + 2(x - 2) = 0$

$x = \frac{5}{4}V,$

$I_2 = \frac{1}{4}A, I_1 = \frac{6}{4}A$

For D



By nodal analysis

$x + x + 1 + 2(x - 2) = 0$

$x = \frac{3}{4},$

$I_2 = \frac{7}{4}A, \ I_1 = \frac{10}{4}A$

25. Ans. (A) →(PQR); (B) →(QST); (C) →(ST); (D) →(PR)

Sol. For (A) : Field remains same as charge is same. Hence energy stored increases. (P) & (Q) as field is same and (R).

For (B) : Since distance between plate and charge remains same and field decreases, (S & T)

$F = \frac{Q^2}{2 \epsilon_0 A}$, charge is same hence force remain same (Q).

For (C) : Since connected to battery P.D. remains same and as distance increases the electric field decreases. (T) Also as capacitance decreases the charge also decreases decreasing the force of interaction and energy inside decrease (S).

For (D) : Capacitance increases and P.D. remaining same, thus charge and energy increases (P) and force of interaction increases. Field remains same as distance and P.D. remains same (R) and energy increases.

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Subjective Type **5 Q. [4 M (0)]**

26. **Ans.** $\frac{\sqrt{4.8}}{2\sqrt{9.1}} \times 10^8$ m/s

27. **Ans.** 4.425×10^{-9} Ampere

Sol. $C_{eq} = \frac{\in_0 (1-x)\times 1}{d} + \frac{k \in_0 (x \times 1)}{d}$

$= \frac{\in_0}{d}[1 - x + kx]$

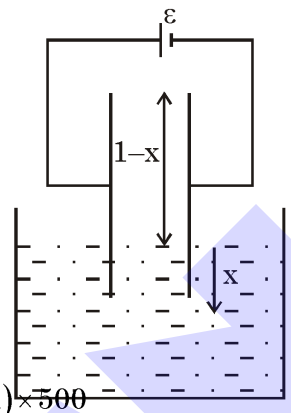
$= \frac{\in_0}{d}[1 + (k-1)x]$

$q = C_{eq}\varepsilon$

$\frac{dq}{dt} = \frac{dC_{eq}}{dt}\varepsilon$

$\Rightarrow \frac{dq}{dt} = \frac{\in_0}{d}(k-1)\varepsilon \Rightarrow I = \frac{(8.85\times 10^{-12})(11-1)\times 500}{0.01}$

$I = 4.425 \times 10^{-9}$ A



28. **Ans.** $\frac{3}{4}$J

Sol. Initial charge on capacitor = CE

Initial potential energy of capacitor = $\frac{CE^2}{2}$

Now $C = \frac{\varepsilon_0 A}{d}$

And new capacitance : $C' = \frac{\varepsilon_0 A}{d/2}$, in series with $\frac{\varepsilon_0 kA}{d/2}$

$\Rightarrow C' = \frac{2\varepsilon_0 A}{d}$ series $\frac{4\varepsilon_0 A}{d}$

$= \frac{\varepsilon_0 A}{d}\left(\frac{2\times 4}{2+4}\right) = \frac{4\varepsilon_0 A}{3d} = \frac{4}{3}C \Rightarrow$ New charge $= \frac{4}{3}CE$

New energy $= \frac{1}{2}\times\frac{4}{3}CE^2 = \frac{2}{3}CE^2$

Now $W_{but} = \Delta H + \Delta U$

$\Rightarrow E\left(\frac{4}{3}CE - CE\right) = \Delta H + \frac{2}{3}CE^2 - \frac{1}{2}CE^2 \Rightarrow \Delta H = \frac{1}{6}CE^2$

29. **Ans.** (a) $\frac{7}{50}$A or $\frac{11}{50}$A, (b) $Q_1 = 9\mu C$, $Q_2 = 0$

Sol. (a) Potential difference across 'C_1' just before swithc is closed.

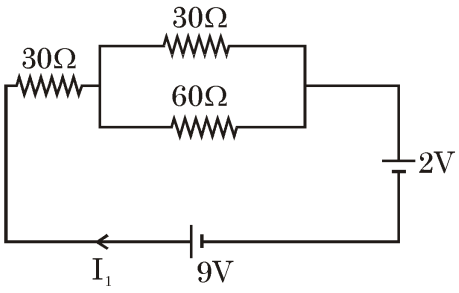
$\Delta V = \frac{q}{C} = \frac{2}{1} = 2V$

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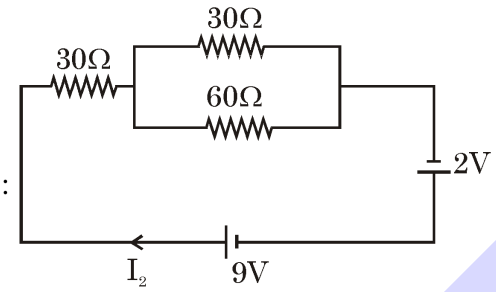
$\therefore$ Just after switch is closed

Case-I :



$$I_1 = \frac{7}{50}A$$

Case-II :



$$I_2 = \frac{9}{50}A$$

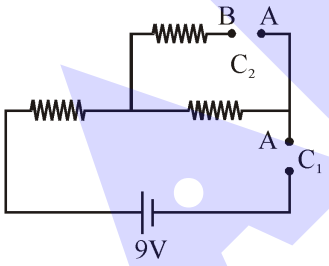
(b) In steady state there will be no current in the circuit

P.d across $C_2 = 0$

So charge = 0

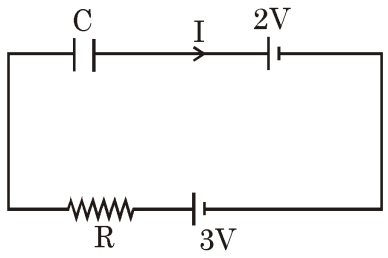
P.d across $C_1 = 9V$

$Q = 9\ \mu C$



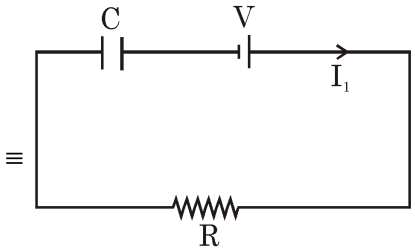
30. Ans. $\frac{V}{R}\left(e^{\frac{-t}{RC}} - 1\right)$

Sol.



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$$I_1 = \frac{V}{R}\left(1 - e^{-\frac{t}{RC}}\right)$$

$$\because I = -I_1 = \frac{V}{R}\left(e^{-\frac{t}{RC}} - 1\right)$$

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